Mean, mode and median

Three different answers to “what is a typical value?” — and the reason a country can have a rising average income and a falling typical one.

LESSON 81–82 | 2 × 40 MIN | ALGEBRA 8, CHAPTER IV §27 | STAGE 9 · 15.4

LESSON CLOCK

5' | 12' | 8' | 13' | 2'

Warm-up	5 min
Explanation	12 min
Interactive	8 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- Calculate the mean, the mode and the median of a list.
- Find the range.
- Estimate the mean of grouped data using midpoints.
- Choose the most suitable average for a given set.

TEXTBOOK

National	Chapter IV, pp. 173–180
Cambridge	Learner’s Book pp. 331–338
Workbook	Workbook 15.4

01 Explanation

The three averages

DEFINITIONS

- **Mean** — add everything and divide by how many: $\bar{x} = \dfrac{\text{sum of values}}{\text{number of values}}$.
- **Mode** — the value that occurs most often. There may be none, or several.
- **Median** — sort the values; the middle one. With an even count, the mean of the two middle ones.

And the **range** = largest − smallest. It is not an average; it measures **spread**.

For $3, 5, 5, 6, 8, 9, 12$: mean $\dfrac{48}{7} \approx 6.9$, mode 5, median 6, range 9.

SORT FIRST, EVERY TIME

The median of $7, 2, 9$ is 7, not 2. Writing the list in order takes five seconds and prevents the commonest error in the topic.

Which average to use

AVERAGE	GOOD BECAUSE	BAD BECAUSE
Mean	uses every value	one extreme value drags it a long way
Median	ignores extremes	ignores most of the data
Mode	the only one for categories	may not exist, or may be at one end

Nine people earn 2 million and one earns 100 million. The **mean** is 11.8 million — a figure nobody earns. The **median** is 2 million, which describes the group honestly. A single **outlier** is exactly what the median is for.

Grouped data

Once the raw values are lost in classes, the mean can only be **estimated**. Assume everything in a class sits at its **midpoint**:

$$\textit{estimated mean} = \frac{\Sigma\,(\text{midpoint} \times \text{frequency})}{\Sigma\ \text{frequency}}$$

CLASS	MIDPOINT	F	MIDPOINT × F
0–10	5	2	10
10–20	15	5	75
20–30	25	8	200
30–40	35	6	210
40–50	45	3	135
Total		**24**	**630**

$$\textit{estimated mean} = \frac{630}{24} = 26.25$$

The **modal class** is the one with the highest frequency — here 20–30.

02 Worked examples

EXAMPLE 1 ***Find the mean, mode, median and range of 4, 7, 3, 7, 9, 2, 7.***

1. $sorted: 2, 3, 4, 7, 7, 7, 9$

   Sort first.

2. $$mean = \frac{39}{7} \approx 5.6$$

3. $mode = 7$

   It appears three times.

4. $median = 7$

   The fourth of seven values.

5. $range = 9 - 2 = 7$

ANSWER

$mean \approx 5.6, mode\ 7, median\ 7, range\ 7$

EXAMPLE 2 ***Find the median of 12, 5, 8, 20, 3, 11.***

1. $sorted: 3, 5, 8, 11, 12, 20$

   Six values — even.

2. The two middle ones are 8 and 11.

3. $$median = \frac{8 + 11}{2} = 9.5$$

   Their mean.

ANSWER

9.5

EXAMPLE 3 **Estimate the mean for classes 0–10, 10–20, 20–30 with frequencies 4, 10, 6.**

1. *midpoints 5, 15, 25*
2. $5 \cdot 4 + 15 \cdot 10 + 25 \cdot 6 = 20 + 150 + 150 = 320$
3. *total frequency* $= 20$
4. $\frac{320}{20} = 16$

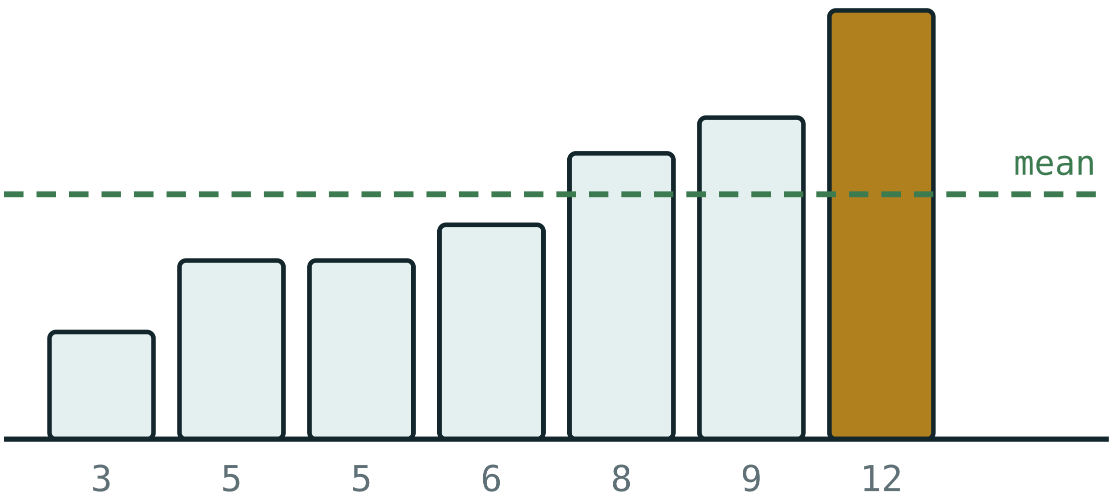
ANSWER

≈ 16

03 Interactive model

Push one value up to 14 and watch the mean move while the median stays still.

Mean, median and mode



values **3, 5, 5, 6, 8, 9, 12** | mean **6.86** | median **6** | mode **5** | range **9**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Mean (average)	O‘rtacha arifmetik	Среднее арифметическое
Mode	Moda	Мода
Median	Mediana	Медиана
Range	Kenglik (farq)	Размах
Sorted order	O‘sish tartibida	В порядке возрастания
Outlier	Chetlangan qiymat	Выброс
Grouped data	Guruhlangan ma’lumotlar	Сгруппированные данные
Midpoint of a class	Sinf o‘rtasi	Середина интервала
Estimate	Baholash	Оценить

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The median of $7, 2, 9$ is:

- 7
- 2
- 9
- 6

One very large value most affects the:

- mode
- median
- mean
- range only

For grouped data the mean is estimated using:

- the class boundaries
- the class midpoints
- the largest value
- the frequencies alone

The range of $3, 5, 5, 12$ is:

- 5
- 9
- 12
- 6.25

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. *Find the mean of 4, 6, 8*

   ANSWER *6*

2. *Find the mode of 2, 3, 3, 5*

   ANSWER *3*

3. *Find the median of 1, 4, 9*

   ANSWER *4*

4. *Find the range of 2, 7, 11*

   ANSWER *9*

5. *Find the mean of 10, 20, 30, 40*

   ANSWER *25*

6. *Find the median of 3, 5, 7, 9*

   ANSWER *6*

7. *Find the mode of 1, 2, 3, 4*

   ANSWER None.

Medium · the standard question

7 PROBLEMS

1. *Find the mean, mode, median and range of 4, 7, 3, 7, 9, 2, 7*

   ANSWER $\approx 5.6, 7, 7, 7$

2. *Find the median of 12, 5, 8, 20, 3, 11*

   ANSWER 9.5

3. *The mean of five numbers is 12. Find their total.*

   ANSWER 60

4. *Four numbers have mean 9. A fifth is added and the mean becomes 10. Find it.*

   ANSWER $50 - 36 = 14$

5. *Estimate the mean for classes 0–10, 10–20, 20–30 with frequencies 4, 10, 6.*

   ANSWER 16

6. *Find the modal class for frequencies 3, 9, 12, 5.*

   ANSWER The third class.

7. *Nine people earn 2 and one earns 100 (millions). Find the mean and median.*

   ANSWER $\text{mean} = 11.8, \text{median} = 2$

Hard · stretch and reason

7 PROBLEMS

1. *The mean of 6 numbers is 15. One number, 20, is removed. Find the new mean.*

   ANSWER $\dfrac{90 - 20}{5} = 14$

2. *A set of 5 numbers has mean 8, median 7 and mode 6. Give a possible set.*

   ANSWER *6, 6, 7, 10, 13* — mean 8.4; adjust to *6, 6, 7, 9, 12* (mean 8) ✓

3. *Estimate the mean for classes 0–20, 20–40, 40–60 with frequencies 5, 12, 3.*

   ANSWER $\dfrac{10{\cdot}5 + 30{\cdot}12 + 50{\cdot}3}{20} = \dfrac{560}{20} = 28$

4. *Why is the median better than the mean for house prices?*

   ANSWER A few very expensive houses drag the mean far above what a typical house costs.

5. *The mean of 10 numbers is 6. Adding an eleventh makes the mean 6.5. Find it.*

   ANSWER $71.5 - 60 = 11.5$

6. *A test has mean 60 for 20 boys and 70 for 30 girls. Find the mean for all 50.*

   ANSWER $\dfrac{1200 + 2100}{50} = 66$

7. *Can the mean, median and mode all be different? Give an example.*

   ANSWER Yes — *1, 2, 2, 5, 10*: mean 4, median 2, mode 2. Try *1, 2, 2, 6, 14*: mean 5, median 2, mode 2.

07 Homework

Homework — 6 problems

Algebra 8, Chapter IV, pp. 173–180. Sort before finding a median.

1. *Find the mean, mode, median and range of 8, 3, 5, 8, 11, 8, 2*
2. *Find the median of 14, 9, 21, 6, 17, 11*
3. *The mean of seven numbers is 11. Find their total.*

4. *Estimate the mean for classes 0–10, 10–20, 20–30, 30–40 with frequencies 3, 8, 6, 3.*
5. *Six numbers have mean 14. A seventh makes the mean 15. Find it.*
6. Explain in two sentences when the median is a better average than the mean.